MATH 345 Skeleton Notes
Unit 3: Beyond linear maps: local linearization
Section 3.4: Partial derivatives, gradients, Hessians, and Jacobian matrices

Partial derivatives

Definition and basic concepts

A partial derivative measures change in one input coordinate while the other input coordinates are held fixed.

Definition: Partial Derivatives Let $f(x, y)$ be a scalar-valued function in two variables. The partial derivative of f with respect to x at (a, b) is the scalar quantity

$$f_x(a, b) = \lim_{h \to 0} \frac{f(a + h, b) - f(a, b)}{h},$$

provided this limit exists. Similarly, the partial derivative of f with respect to y at (a, b) is the quantity

$$f_y(a, b) = \lim_{h \to 0} \frac{f(a, b + h) - f(a, b)}{h}$$

provided this limit exists.

We compute the partial derivative in x of an expression precisely by treating the y-variable as a constant, and differentiating in x. Similarly, we compute the partial derivative in y by treating the x-variable as a constant.

Note There are several notations used for the partial derivative of a function $f(x, y)$, namely, we might write $f_x(a, b)$ as

$$\partial_x f(a,b) \quad \frac{\partial f}{\partial x}(a,b) \quad (D_x f)(a,b) \quad \text{or} \quad (D_1 f)(a,b).$$

Similarly, we write $f_y(a, b)$ as

$$\partial_y f(a,b) \quad \frac{\partial f}{\partial y}(a,b) \quad (D_y f)(a,b) \quad \text{or} \quad (D_2 f)(a,b).$$

Activity

Let

$$f(x, y) = x^2 - 3xy + 2y^2 - 4x + 5y - 12$$

Calculate $f_x(x, y)$ directly using the definition of the derivative.

Activity

Let

$$f(x, y) = x^2 - 3xy + 2y^2 - 4x + 5y - 12$$

Calculate $f_y(x, y)$ directly using the definition of the derivative.

Activity

For each function $f(x, y)$, find $f_x(x, y)$ and $f_y(x, y)$.
$f(x, y) = 3x^2y + 2x - 3y$

Activity

For each function $f(x, y)$, find $f_x(x, y)$ and $f_y(x, y)$.
f(x,y) = x\sin y + y\sin x

Activity

For each function $f(x, y)$, find $f_x(x, y)$ and $f_y(x, y)$.
$f(x, y) = x^2e^{xy}$.

Geometric interpretation

Definition: The Slice of a Function Given a function $f(x, y)$ and $b \in \mathbb{R}$, the slice of the function f in the plane $y = b$ is the function $g : \mathbb{R} \to \mathbb{R}$ given by $g(x) = f(x, b)$. Similarly, for $x \in \mathbb{R}$ the slice of f in the plane $x = a$ is the function $g : \mathbb{R} \to \mathbb{R}$ given by $g(y) = f(a, y)$.

Geometrically, the slice of a function f in the plane $y = b$ corresponding to intersecting the surface given by the equation $z = f(x, y)$ with the plane $y = b$, creating a curve in this plane.

Activity

Sketch the slices of the function $f(x, y) = \sin(x + y^2)$ in the plane $y = b$.

Theorem: Geometric Interpretation If we consider the surface in $\mathbb{R}^3$ described by the equation $z = f(x, y)$, then

- $f_x(a, b)$ is the *slope of the tangent line* to the curve formed by intersecting the surface $z = f(x, y)$ with the vertical plane $y = b$.
- $f_y(a, b)$ is the *slope of the tangent line* to the curve formed by intersecting the surface $z = f(x, y)$ with the vertical plane $x = a$.

Activity

Let $f(x, y) = 16 - x^2 - y^2$. Find $f_x(1, 2)$ and $f_y(1, 2)$ and interpret these numbers as slopes of a certain curve.

Activity

Consider the contour plot shown in textbook figure fig-8-5-contour-plot-example-increment-point2 for a function F with contour lines at increments of 0.2. In particular, consider the curve obtained from the slice $y = 0$, indicated by the red line in textbook figure fig-8-5-contour-plot-example-increment-point2. By interpreting the contour plot, what can we say about the slopes of this curve, or equivalently, the values of F_x on this line?

Activity

Consider textbook figure fig-8-5-contour-ex-2, which is the same contour plot as in textbook figure fig-8-5-contour-plot-example-increment-point2, but with the vertical line $x = 2.25$ indicated instead. As in the activity, read off the behaviour of the partial derivatives F_y from the contour plot.

Generally, the spacing of contour lines corresponds to the magnitude of partial derivatives, and the sign of the partial derivatives corresponds to whether the function increases or decreasing in that direction.

Activity

Consider the function $f(x, y) = x(y^2 + 1)$, whose contour plot is indicated in textbook figure fig-8-5-contour-ex-3. Five points are marked on the plot: $P = (0, 0)$, $Q = (3/2, 0)$, $R = (7/2, 0)$, $S = (2, 1)$, and $T = (2, -1)$. For each of these five points, determine from the contour plot whether f_x is positive, negative, or zero, and whether f_y is positive, negative, or zero. Then calculate these derivatives exactly and check whether your observations were correct.

Partial derivatives with more variables

The concept of partial derivatives extends naturally to functions of more than two variables.

Definition For a function $f(\mathbf{x}) = f(x_1, \dots, x_n)$ of n variables, the partial derivative with respect to x_i is defined as

$$f_{x_i}(\mathbf{x}) = \lim_{h \to 0} \frac{f(x_1, \dots, x_i + h, \dots, x_n) - f(x_1, \dots, x_n)}{h}$$

provided this limit exists. Using vector notation, we can write this definition more cleanly, by writing

$$f_{x_i}(\mathbf{x}) = \lim_{h \to 0} \frac{f(\mathbf{x} + h\mathbf{e}_i) - f(\mathbf{x})}{h},$$

where $\mathbf{e}_1, \dots, \mathbf{e}_n$ are the standard basis vectors (recall Standard basis). This partial derivative is also written as

$$\frac{\partial f}{\partial x_i} \quad \partial_{x_i} f \quad \text{or} \quad (D_i f)(\mathbf{x}).$$

If the variables $x_1, \dots, x_n$ are denoted using different symbols, these symbols might also be used to denote the partial derivatives instead. For instance, in $\mathbb{R}^3$, we might write a function as $f(x, y, z)$, and then the three partial derivatives of f may be denoted using any of the following four notations:

- $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$, and $\frac{\partial f}{\partial z}$.
- f_x, f_y, and f_z.
- $\partial_x f, \partial_y f$ and $\partial_z f$.
- $D_1 f$, $D_2 f$, and $D_3 f$.

As for functions of two variables, we can calculate these derivatives by treating all variables but that we are taking the partial derivative of as constants.

Activity

If $f(x, y, z) = 5x^2y^2z^3$, find f_x, f_y, and f_z.

Activity

Find f_x, f_y, and f_z where $f(x, y, z) = xy + yz + xz$.

Activity

Find f_x and f_y where $z = f(x, y) = x \ln y + ye^x$.

Higher-order partial derivatives

Given a scalar-valued function f, the partial derivatives of f (where defined) are also scalar-valued functions, so we can consider partial derivatives of that function as well, which leads to *higher order partial derivatives*.

Definition If f is a function of two variables, then the second-order partial derivatives of f are defined as:

- The derivative f_{xx} or $\frac{\partial^2 f}{\partial x^2}$, obtained by taking the partial derivative of f twice in the x variable.

- The derivative f_{yx} or $\frac{\partial^2 f}{\partial y \partial x}$, obtained by *first* differentiating f in the x variable, and *then* differentiating f in the y variable.

- The derivative f_{xy}, or $\frac{\partial^2 f}{\partial x \partial y}$, obtained by *first* differentiating f in the y variable, and *then* differentiating f in the x variable.

- The derivative f_{yy}, or $\frac{\partial^2 f}{\partial y^2}$, obtained by differentiating f twice in the y variable.

Similarly, for a function $f(x_1, \ldots, x_n)$, the higher derivatives of f are defined as $f_{x_i x_j} = \frac{\partial^2 f}{\partial x_i \partial x_j}$ for $1 \leq i, j \leq n$, obtained by first differentiating f in the variable x_j, and then differentiating f in the x_i variable.

Theorem: Clairaut's Theorem Suppose a scalar-valued function f is defined in an open ball B that contains a point $\mathbf{a}$. If the functions $f_{x_i x_j}$ and $f_{x_j x_i}$ are well-defined and continuous on the ball B, then $f_{x_i x_j}(\mathbf{a}) = f_{x_j x_i}(\mathbf{a})$.

Activity

In this activity we compute all four partial derivatives of the function $f(x, y) = x \ln y + ye^x$.
First compute f_x and f_y.

Activity

In this activity we compute all four partial derivatives of the function $f(x, y) = x \ln y + ye^x$.
Next, compute f_{xx} and f_{yy}.

Activity

In this activity we compute all four partial derivatives of the function $f(x, y) = x \ln y + ye^x$.
Now compute f_{xy} and f_{yx}.

Note For most functions f encountered in practice, the second-order partial derivatives are continuous where defined, and so Clairaut's Theorem applies. However, there are functions for which mixed partial derivatives are not equal to one another.

Activity

Find all four second partial derivatives of the function

$$f(x, y) = \ln(x^3 + y^2).$$

Gradients and Hessians

The gradient collects first partial derivatives of a scalar-valued function.

Definition: Gradient Let f be a scalar-valued function of n variables. At a point $\mathbf{x}$ where all first partial derivatives of f exist, the gradient of f is the vector

$$(\nabla f)(\mathbf{x}) = \begin{bmatrix} D_1 f(\mathbf{x}) \\ \vdots \\ D_n f(\mathbf{x}) \end{bmatrix}.$$

When these vectors are defined throughout a domain, they form a vector-valued function ∇f.

Activity

Let $f(x, y) = \sin(x + y^2)$. Find the gradient of f.

Activity

Let $f(x, y) = x^2 + xy + y^2$. Compute $\nabla f(1, 2)$.

An important property of the gradient of a function at a point is that it is a vector that is perpendicular to the level curves of the function.

Theorem Let $f(x, y)$ be a function with continuous partial derivatives.

If (a,b) is a point where $\nabla f(a,b) \neq \mathbf{0}$, then $\nabla f(a,b)$ is $\boxed{\text{perpendicular to the level curve}}$ of f passing through the point (a,b).

See textbook figure fig-8-5-gradient-level-curves for a graphical example of the theorem.
The Hessian collects second partial derivatives.

Definition Let f be a scalar-valued function of n variables, and suppose the second partial derivatives of f exist at a point $\mathbf{x}$. The $\boxed{\text{Hessian matrix}}$ of f at $\mathbf{x}$ is the $n \times n$ matrix of second partial derivatives:

$$H_f(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2}(\mathbf{x}) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(\mathbf{x}) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(\mathbf{x}) & \frac{\partial^2 f}{\partial x_2^2}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n}(\mathbf{x}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(\mathbf{x}) & \frac{\partial^2 f}{\partial x_n \partial x_2}(\mathbf{x}) & \cdots & \frac{\partial^2 f}{\partial x_n^2}(\mathbf{x}) \end{bmatrix}$$

That is, $H_f(\mathbf{x})$ is the matrix whose (i,j)-entry is the second derivative $\frac{\partial^2 f}{\partial x_i \partial x_j}$ obtained by first differentiating in the x_j variable, and then the x_i variable. When the relevant mixed second partial derivatives are continuous on a ball around $\mathbf{x}$, Clairaut's Theorem implies that the Hessian is symmetric at $\mathbf{x}$.

In particular, the Hessian of a scalar-valued function of two variables can be written

$$H_f = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix} = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}.$$

Activity

Find the Hessian matrix for the function

$$f(x, y) = \sin(x + y^2)$$

at a general point (x, y).

Activity

Find the Hessian matrix of the function

$$f(x, y) = x^2 + xy + y^2$$

at a general point (x, y)

Activity

Let $f(x, y, z) = xy + yz + xz$. Compute its gradient ∇f and Hessian H_f at a general point (x, y, z).

Jacobian matrices

For vector-valued maps, the Jacobian matrix collects the partial derivatives of all component functions.
The rows correspond to component functions. The columns describe how the output changes when one input coordinate changes.

Definition: Jacobian matrix For a vector-valued map $F : \mathbb{R}^n \to \mathbb{R}^m$ given by

$$F(\mathbf{x}) = \begin{bmatrix} F_1(\mathbf{x}) \\ F_2(\mathbf{x}) \\ \vdots \\ F_m(\mathbf{x}) \end{bmatrix}$$

where $\mathbf{x} = (x_1, x_2, \dots, x_n)$, the

Jacobian matrix of F is the $m \times n$ matrix:

$$J_F(\mathbf{x}) = \begin{bmatrix} \frac{\partial F_1}{\partial x_1} & \frac{\partial F_1}{\partial x_2} & \cdots & \frac{\partial F_1}{\partial x_n} \\ \frac{\partial F_2}{\partial x_1} & \frac{\partial F_2}{\partial x_2} & \cdots & \frac{\partial F_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial x_1} & \frac{\partial F_m}{\partial x_2} & \cdots & \frac{\partial F_m}{\partial x_n} \end{bmatrix}$$

The Jacobian matrix generalizes the concept of the derivative to a vector-valued map of multiple variables. It provides a compact representation of all first-order partial derivatives of a function.

Activity

Find the Jacobian matrix of $f(x, y, z) = xy + yz + xz$.

Activity

Compute the Jacobian matrix of $F(x, y) = (x^2 - y^2, 2xy, x + y)$.

Activity

Compute the Jacobian matrix of the gradient function $\nabla f(x, y)$ for $f(x, y) = x^2 + xy + y^2$.